CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The height of a parallelogram of area 184 cm^2 and base 23 cm is (cm): 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

2. A square has area 324 cm^2. Its perimeter (cm) is: 1

(a) 72 **(b)** 73 **(c)** 74 **(d)** 71

3. The area of a quadrilateral whose diagonal is 18 cm and whose offsets from the diagonal are 7 cm and 5 cm is (cm^2): 1

(a) 107 **(b)** 109 **(c)** 110 **(d)** 108

4. A rectangle of perimeter 62 cm has length 22 cm. Its area (cm^2) is: 1

(a) 199 **(b)** 198 **(c)** 200 **(d)** 197

5. A circular wire of radius 21 cm (take $\pi = \frac{22}{7}$) is bent into a square. The side of the square (cm) is: 1

(a) 33 **(b)** 35 **(c)** 34 **(d)** 32

6. A wheel of diameter 56 cm (take $\pi = \frac{22}{7}$) makes 25 revolutions. The distance covered (in metres) is: 1

(a) 44 **(b)** 43 **(c)** 46 **(d)** 45

7. The length of the arc of a sector of angle 36° in a circle of radius 35 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 23 **(b)** 24 **(c)** 22 **(d)** 21

8. A triangle has area 168 cm^2 and one side 14 cm. The altitude on this side (cm) is: 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 23

9. The height of a parallelogram of area 266 cm^2 and base 19 cm is (cm): 1

(a) 13 **(b)** 15 **(c)** 16 **(d)** 14

10. A square has area 576 cm^2. Its perimeter (cm) is: 1

(a) 98 **(b)** 95 **(c)** 97 **(d)** 96

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a quadrilateral $ABCD$ in which diagonal $AC = 30$ cm and the perpendiculars from B and D to AC are 8 cm and 12 cm. 2

12. Find the arc length for 72° in a circle of radius 35 cm (take $\pi = \frac{22}{7}$). 2

13. A triangle has sides 15 cm, 15 cm and 24 cm. Find its area and the altitude on the side 24 cm. 2

14. Find the area of a quadrilateral $ABCD$ in which diagonal $AC = 22$ cm and the perpendiculars from B and D to AC are 10 cm and 6 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the cyclic quadrilateral with sides 4 cm, 10 cm, 20 cm and 22 cm. 3

16. A rectangular sheet 40 cm $\times$ 14 cm has a semicircle of diameter 14 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

17. The sides of a parallelogram are 20 cm and 16 cm and the height on the side 20 cm is 12 cm. Find its area and the height on the side 16 cm. 3

OR

Find the area of the cyclic quadrilateral with sides 5 cm, 12 cm, 21 cm and 22 cm. 3

18. A rectangular sheet 35 cm $\times$ 28 cm has a semicircle of diameter 28 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

OR

The sides of a parallelogram are 24 cm and 18 cm and the height on the side 24 cm is 9 cm. Find its area and the height on the side 18 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular plot 60 m $\times$ 35 m has a circular pond of radius 14 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹5 per m^2. 5

OR

A triangular field has sides 25 m, 25 m and 14 m. Find its area, and the cost of ploughing it at ₹3 per m^2. 5

20. A rectangular plot 45 m $\times$ 45 m has a circular pond of radius 7 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹8 per m^2. 5

OR

A triangular field has sides 17 m, 25 m and 26 m. Find its area, and the cost of ploughing it at ₹2 per m^2. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND